Network Devices (Hub, Repeater, Bridge, Switch, Router, Gateways and Brouter)

Network devices or networking hardware are the physical devices that are used for establishing connections and facilating interaction between different devices in a computer network.

Hub

Hubs work in the physical layer of the OSI model. A hub is a device for connecting multiple Ethernet devices and making them act as a single network segment. It has multiple inputs and output ports in which a signal introduced at the input of any port appears at the output of every port except the original incoming port.

A hub can be used with both digital and analog data. Hubs do not perform packet filtering or addressing function, they send the data packets to all the connected devices.

Types of Hub −

- Active Hub
- Passive Hub
- Intelligent Hub

Repeater

A repeater operates at the physical layer of the OSI model.

- A Repeater connects two segments of a network cable.
- Sometimes it regenerates the signals to proper amplitudes and sends them to the other segment.
- If the signal becomes weak, it can copy the signal bit by bit and regenerate it at the original strength.
- It is a 2-port device.

Bridge

A bridge operates at the data link layer of the OSI model. It can read only the outmost hardware address of the packet but cannot read the IP address. It reads the outmost section of the data packet to tell where the message is going. It reduces the traffic on other network segments. It does not send all the packets. So, a bridge can be programmed to reject packets from a particular network.

Switch

Switches may operate at one or more layers of the OSI model. They may operate in the data link layer and network layer; a device that operates simultaneously at more than one of these layers is known as a *multilayer switch*.

A Switch can check the errors before forwarding the data, which makes it more efficient and improves its performance. A switch is the better version of a hub. It is a multi-port bridge device.

Router

Routers are small physical devices that operate at the network layer to join multiple networks together.

- A router is a device like a switch that routes data packets based on their IP addresses.
- Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets.
- A Router divides the broadcast domains of hosts connected through it.
- Routers perform the traffic directing functions on the Internet. A data packet is typically forwarded from one router to another through the networks that constitute the internetwork until it reaches its destination code.
- Routers may also be used to connect two or more logical groups of computer devices known as subnets, each with a different subnetwork address. The subnet addresses recorded in a router do not necessarily map directly to the physical interface connections.

Two types of routers −

- **Static routers** – Static routers are configured manually and route data packets based on the information in a router table.
- **Dynamic routers** – Dynamic routers use adaptive routing which is a process where a router can forward data by a different route.

Gateway

A gateway is an internetworking capable of joining together two networks that use different base protocols. A network gateway can be implemented completely in software, hardware, or a combination of both, depending on the types of protocols they support.

A network gateway can operate at any level of the OSI model. A broadband router typically serves as the network gateway, although ordinary computers can also be configured to perform equivalent functions.

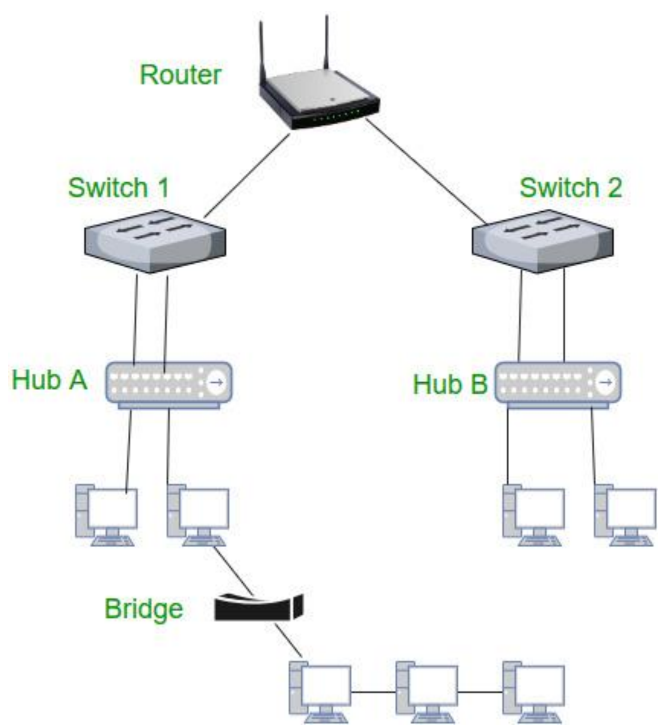
- A gateway is a router or proxy server that routes between networks.
- A gateway belongs to the same subnet to which the PC belongs.

Brouter

A **brouter** is a combination of a Bridge and a Router. It provides the functions of a bridge and a router, so it can operate at the data link and network layers of the OSI Model.

- A Brouter can connect networks that use different protocols.
- It can be programmed to work only as a bridge or only as a router.
- When it is configured as a bridge, it forwards data packets to the appropriate segment using a specific protocol.
- When it is configured as a router, it routes the data packets to the appropriate network using a routed protocol such as IP.

Most brouters are simply routers that have been configured. The main function of a bridge is to connect two different LAN segments using the same protocol. A router, on the other hand, is used to route the packets in a network.



NIC – NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC card is a layer 2 device which means that it works on both the physical and data link layers of the network model.